

An Internship Report
Of
Artificial Intelligence Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE)



By

Roll Number of Student: 25SCS1003005166

Name of Student: RAMESH KUMAR

Batch: 2026-27

CANDIDATE'S DECLARATION

I, RAMESH KUMAR, do hereby declare that the internship report titled “**Artificial Intelligence Internship**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Student Name: **RAMESH KUMAR**

Roll No.: **25SCS1003005166**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature:

Student Name: **RAMESH KUMAR**

Roll No.: **25SCS1003005166**

TABLE OF CONTENTS

1. Candidate's Declaration	i
2. Acknowledgement	ii
3. Internship Completion Certificate.....	-
4. Project Description.....	1
4.1 Introduction.....	1
4.2 Organization Profile.....	2
4.3 Problem Statement.....	3
4.4 Project Objectives	3
4.5 Scope of the Project	4
4.6 Technologies and Tools Used.....	4
4.7 System Architecture.....	5
4.8 Methodology	6
4.9 Expected Outcomes	7
4.10 Certificates of Completion and Communication Proof	8
5. Bibliography/References.....	9

3. INTERNSHIP COMPLETION CERTIFICATE



4. PROJECT DESCRIPTION

4.1 Introduction

The Artificial Intelligence Internship was undertaken to develop a fundamental understanding of Artificial Intelligence, Machine Learning, and their practical applications. The internship followed a structured curriculum covering Python for AI development, data preprocessing and visualization, supervised learning, model evaluation, neural networks, Convolutional Neural Networks (CNNs), Natural Language Processing (NLP), and Reinforcement Learning.

The theoretical concepts were complemented by practical examples and implementations, providing an understanding of how AI techniques can be applied to real-world problems. As part of the practical work, a Content-Based Movie Recommendation System was developed using Python and machine learning techniques. The project enabled the application of concepts such as data handling, feature extraction, NLP, and similarity-based recommendation in a practical setting.

4.2 Organization Profile

Codec Technologies is an IT consultancy, technology services, and educational training organization that provides a range of technology-related solutions. Its services include web and application development, cloud hosting, software deployment, infrastructure management, and digital transformation.

The organization also offers corporate training and internship programs across various technical domains, including Artificial Intelligence, Data Science, Python Development, Cyber Security, Web Development, and Data Analytics. These programs are structured to provide learners with theoretical knowledge as well as practical exposure to industry-relevant technologies and applications.

4.3 Problem Statement

Artificial Intelligence is a broad field that encompasses various techniques, including Machine Learning, Deep Learning, Natural Language Processing, and Reinforcement Learning. Developing an understanding of these concepts requires both theoretical knowledge and practical experience in areas such as data processing, model development, and evaluation.

The internship aimed to provide a structured introduction to the fundamental concepts of AI and demonstrate their practical applications. It focused on understanding the workflow of AI-based systems, from data preparation and feature extraction to model development and performance evaluation.

As part of the practical component, a Content-Based Movie Recommendation System was developed to recommend movies similar to a user-selected movie based on its content and available metadata.

4.4 Project Objectives

- To understand the fundamental concepts and applications of Artificial Intelligence.
- To develop basic programming and data-handling skills using Python.
- To learn techniques for data preprocessing, visualization, and preparation for AI applications.
- To understand the fundamentals of supervised learning and machine learning models.
- To learn the basics of model evaluation and performance measurement.
- To gain an introductory understanding of neural networks and deep learning.
- To understand the working and applications of Convolutional Neural Networks (CNNs) for image-related tasks.
- To explore Natural Language Processing and basic text-processing techniques.
- To understand the basic principles and applications of Reinforcement Learning and Q-learning.
- To apply the concepts learned during the internship through a practical AI project.

4.5 Scope of the Project

The scope of the internship covers the fundamental areas required to develop a basic understanding of Artificial Intelligence and its applications. It includes Python-based data handling, data preprocessing, supervised machine learning, model evaluation, neural networks, CNNs, Natural Language Processing, and reinforcement learning.

The practical scope involves implementing AI and Machine Learning concepts through various examples and a project. The Content-Based Movie Recommendation System demonstrates the application of data processing, feature extraction, and similarity calculation to generate relevant movie recommendations based on their content.

The knowledge gained during the internship provides a foundation for further exploration of advanced AI technologies. Future improvements may include the use of larger datasets, advanced deep learning and NLP techniques, hybrid

recommendation approaches, and deployment of AI applications through web or mobile platforms.

4.6 Technologies and Tools Used

- **Python** – Used for programming and implementation of AI and machine learning concepts.
- **NumPy** – Used for numerical operations and array-based data processing.
- **Pandas** – Used for data manipulation, cleaning, and analysis.
- **Matplotlib** – Used for data visualization.
- **Scikit-learn** – Used for machine learning, feature extraction, similarity calculation, and evaluation.
- **NLTK and SpaCy** – Introduced for Natural Language Processing and text-processing tasks.
- **Jupyter Notebook / Google Colab** – Used for writing, executing, and documenting Python-based implementations.
- **Machine Learning techniques** – Used for understanding supervised learning and model evaluation.
- **Neural Networks and CNNs** – Studied as fundamental deep learning techniques.
- **Q-learning** – Studied as a basic Reinforcement Learning approach.

4.7 System Architecture

The overall internship covered multiple Artificial Intelligence concepts and therefore did not follow a single system architecture. However, the practical project developed during the internship followed a structured data-processing and recommendation pipeline.

The architecture of the Content-Based Movie Recommendation System starts with a movie dataset containing information such as titles, genres, descriptions, keywords, and other metadata. The data is loaded and preprocessed to handle missing or irrelevant information. Relevant features are then selected and combined to create a suitable representation of each movie.

These features are converted into numerical vectors, allowing the system to compare movies mathematically. Cosine similarity is then used to determine the similarity between movies, and the resulting scores are used to rank them. Finally, the movies with the highest similarity scores are selected and displayed as recommendations.

4.8 Methodology

The internship followed a structured and progressive learning approach, starting with the fundamentals of Artificial Intelligence and Python programming and gradually introducing more advanced concepts. The initial stage focused on understanding AI fundamentals, Python-based data handling, data preprocessing, and visualization.

The next stage covered the basics of Machine Learning, including supervised learning and model evaluation techniques. This was followed by an introduction to neural networks and Deep Learning, with a focus on Convolutional Neural Networks (CNNs) and their applications in image classification. The final stage introduced Natural Language Processing, text processing, sentiment analysis, and Reinforcement Learning using basic Q-learning techniques.

Practical examples and implementations were used throughout the internship to reinforce the concepts covered in the learning modules. As part of the project, a movie dataset was processed using Python, and relevant features were extracted and transformed into numerical representations. Cosine similarity was then applied to identify movies with similar content, and the resulting similarity scores were used to generate a ranked list of recommendations.

4.9 Expected Outcomes

The internship is expected to provide a strong foundational understanding of Artificial Intelligence and its major areas of application. It enables the learner to develop familiarity with Python-based data handling, data preprocessing, Machine Learning, model evaluation, neural networks, CNNs, Natural Language Processing, and Reinforcement Learning.

The internship also provides practical experience in applying AI concepts to real-world problems. Through the development of the Content-Based Movie Recommendation System, practical skills are developed in handling datasets, processing textual information, extracting relevant features, calculating similarity, and generating movie recommendations.

Overall, the internship provides a foundation for further learning and development in areas such as Machine Learning, Deep Learning, NLP, Computer Vision, and AI-based application development.

4.10 Certificates of Completion and Communication Proof



5. BIBLIOGRAPHY/REFERENCES

Bibliography / References

- Codec Technologies, *Artificial Intelligence Internship – Course Modules and Learning Materials*.
- Python Software Foundation, *Python 3 Documentation*.
- NumPy Developers, *NumPy Documentation and User Guide*.
- Pandas Development Team, *Pandas Documentation and User Guide*.
- Scikit-learn Developers, *Scikit-learn User Guide and API Reference*.
- TensorFlow, *TensorFlow Documentation*, reference for neural networks, deep learning, and CNN concepts.
- Natural Language Toolkit (NLTK), *NLTK Documentation*, reference for Natural Language Processing and text-processing techniques.
- spaCy, *spaCy Documentation*, reference for NLP and linguistic data processing.
- Goodfellow, I., Bengio, Y., and Courville, A., *Deep Learning*, MIT Press.
- Russell, S. and Norvig, P., *Artificial Intelligence: A Modern Approach*, Pearson.
- .